

V2X-SIEM: A Lightweight Security Information and Event Management Architecture for Connected and Autonomous Vehicles

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Abstract—The growing complexity of vehicular networks, from the implementation of Vehicle-to-Everything (V2X) communication technologies, poses a significant challenge to traditional Security Information and Event Management (SIEM) systems, which are often inflexible and unable to support timely threat detection in automotive contexts. This paper proposes a modular, lightweight, and open-source architecture for a SIEM system designed for Connected and Autonomous vehicles (CAVs). The proposed architecture is built on Elasticsearch and Kibana, and is configured for low latency, high scalability, and efficient event correlation from vehicular communication protocols. It includes components for event collection, analysis, and visualization, and processes vehicular security events nearly in real time while remaining resource efficient, which makes it suitable for deployment with edge computing. The contributions are demonstrated to have practical implications for vehicle security by bridging the strategies of enterprise SIEM capabilities and the unique demands of intelligent transportation systems.

Keywords—Security Information and Event Management (SIEM), Vehicular Networks, C-V2X, DSRC, Elastic Security

I. INTRODUCTION

The increasing connectivity of modern vehicles, driven by technologies such as Dedicated Short-Range Communications (DSRC) and Cellular Vehicle-to-Everything (C-V2X), has introduced novel security challenges in the automotive domain. As vehicles exchange safety-critical data with each other, the infrastructure, and the network, the need for real-time monitoring and threat detection becomes crucial. Beyond safety concerns, the economic value of vehicular data is also increasing rapidly. A recent report by Otonomo estimated that the automotive data market, valued at \$101 billion in 2021, is projected to double and reach \$204 billion by 2030 [1]. Despite

this growing importance, traditional Security Information and Event Management (SIEM) systems, widely used in IT-specific environments, are not designed to manage the requirements of vehicular networks. Typically, SIEM systems process logs from various network components like routers and firewalls, correlating them to detect potential security threats [2]. However, as networks grow, these systems become harder to manage and more expensive. The core capabilities of SIEMs were discussed in a recent study [3], emphasizing the importance of real-time data processing, scalability, and integration with diverse data sources, which are critical to handle the heterogeneous data coming from vehicular data. They highlighted the limitations of traditional SIEMs, such as basic correlation rules and weak data visualization, which are more challenging in dynamic, mobile environments such as vehicular networks. In the automotive domain, where mobility and low latency are critical, these traditional systems fail to address the complexity of vehicular data traffic. This paper presents a SIEM framework tailored for automotive protocols, capable of processing and analyzing events from DSRC and C-V2X communications. Our approach aims to address the unique challenges of vehicular networks, such as mobility and low latency, while offering reliable security monitoring.

II. RELATED WORKS

The implementation of SIEM systems has been widely explored with both commercial products like ArcSight and Splunk and open-source alternatives like the ELK Stack (Elasticsearch, Logstash, Kibana) and Wazuh, with trade-offs in flexibility, cost, and integration. Beyond the specific use cases of the ELK Stack, additional research studies have compared SIEM platforms in terms of their compliance and scalability in heterogeneous environments. K et al. presented an evaluation

framework that compared several of the most used SIEM systems [4]. Their survey highlighted challenges in SIEM adoption, such as complex setup processes, data overload, integration issues, and high implementation costs, emphasizing the complexity of selecting and implementing an appropriate SIEM, considering both its technical and operational requirements. Similarly, another study [5] implemented a SIEM using Splunk Enterprise and deployed it on a local network with components such as the Indexer, used for processing and storing log data; the Search Head, for supporting querying and reporting; and the Forwarders, for collecting and forwarding information to the Indexer. Their system also included OSINT-based IP reputation lookups, which reaffirmed the value of combining rule-based detection supported by third-party threat intelligence. ArcSight was used in the study [6] to introduce a customized incident lifecycle framework, since the authors acknowledged that existing models (MITRE ATT&CK, Cyber Kill Chain, and Mandiant's lifecycle) are not fully comprehensive. This demonstrated that tailoring detection rules can reduce noise and increase the accuracy.

Open-source solutions, especially Wazuh, have gained attention for endpoint monitoring and integration with other threat detection tools. In [7], Wazuh was integrated with a custom Malware Analysis Hub that included Yara, Capa, and VirusTotal. The Malware Analysis Hub, developed with Flask, was composed of three modules: File Insights, URL Scanner, and Network Traffic Monitor. The system automatically triggered malware analysis events and displayed real-time data on the Wazuh dashboard, demonstrating a comprehensive and centralized endpoint protection approach.

In [8], the authors compared three open-source SIEM tools—Wazuh, the ELK Stack, and OSSIM—and concluded that Wazuh stands out with its superior detection capabilities. They deployed Wazuh on three virtual machines and enabled modules for system monitoring, file integrity checks, threat detection, and automated response. Simulated cyberattacks were used to evaluate the system's capabilities, confirming Wazuh's effectiveness in alert visualization, attack tactic mapping using the MITRE ATT&CK framework, and supporting threat response in real time.

Similarly, the paper [9] also evaluated Wazuh detection and response with two use cases: DoS attack detection and file integrity monitoring (FIM). Their environment included Wazuh agents integrated with Suricata or Snort, while the attack traffic originated from Kali Linux and Windows machines. Wazuh demonstrated effective threat detection and response in both use cases. For FIM, Wazuh efficiently monitored and alerted about unauthorized file changes, and its active response capabilities allowed for the potential of automated mitigation actions such as blocking attacker's IP using firewall rules.

While the previous paper focused on Wazuh's standalone effectiveness in endpoint protection, the research [10] extended on this approach by integrating Wazuh with multiple intrusion detection systems—Snort, Suricata, Zeek—to further enhance DDoS detection. These platforms monitored and examined network traffic in their system, and Wazuh acted as the central SIEM correlating and displaying the alerts in its dashboard. The experiment results showed that Snort provided the highest

detection accuracy, while Wazuh aggregated the outputs with minimal system overhead. Despite the system's success, the authors noted that fine-tuning alert threshold remained a challenge to improve precision.

Apart from Wazuh-based implementations, the ELK Stack has been used in sectors such as healthcare, critical infrastructure, smart transportation, and smart cities, as a central component of SIEM implementations for its ability to handle logs in real time and its advanced data visualization. In healthcare, the study [11] combined the ELK Stack with open-source technologies such as Apache Flink, Apache Storm, Kafka to enable real-time threat identification and analysis. Homomorphic encryption and Trusted Execution Environments (TEE) ensured data confidentiality, enabling computation over encrypted data while protecting sensitive patient information. In another domain, the research [12] developed the SPEAR SIEM framework for smart grids, integrating the ELK Stack with OSSIM, Kafka, and a Visual-based IDS (VIDS) to identify and correlate threats in the industrial control systems of smart grids. For smart cities environments, the research presented in [13] proposed a SIEM-as-a-Service (SOCaaS-IoT) solution using ELK, Wazuh, and TheHive, allowing scalable monitoring for IoT devices in a smart city context.

Further exploring the usability of ELK towards large-scale implementation, the paper in [14] developed a Dockerized Elastic Stack with Docker Compose to orchestrate containers for Elasticsearch, Logstash, Kibana, and Beats. During experiments, the authors measured the comparison of performance between a dockerized setup against a standalone ELK configuration. Results showed that the dockerized structure was more efficient, offering more ease of use, and yielded minimal performance trade-offs, affirming the practical benefits of containerization in large-scale SIEM implementation.

Finally, the paper in [15] extended the concept of scalable and modular SIEMs with their GLACIER architecture that used Elasticsearch and Kibana for event storage and analysis. Their machine learning-based anomaly detection system also used Docker containers and RabbitMQ for communication. Tested in a real-world enterprise environment, the ability of the system to detect traffic anomalies was validated, although the authors noted that additional data would be necessary to improve the detection accuracy in high-traffic environments.

III. PROBLEM FORMULATION

While prior research illustrated the versatility and effectiveness of SIEM platforms in areas such as healthcare, smart grids, and IoT applications, there is a lack of literature that directly addresses lightweight and scalable SIEM solutions for connected vehicles. This paper aims to bridge the gap in the literature by presenting a SIEM system tailored to the vehicular domain, with a focus on modularity and resource efficiency.

The recent proliferation of the Internet of Vehicles and its associated subsystem, Connected and Autonomous Vehicles, has led to a greater volume and complexity of data being generated by vehicle systems. Despite this evolution, the current SIEM solutions remain primarily enterprise-oriented, lacking the adaptability required for particular environments such as the

automotive domain. Typical SIEMs are not designed to aggregate vehicular data protocols such as DSRC or C-V2X, making them impractical for real-time threat detection in vehicles.

According to Gartner Peer Insights on "Security Information and Event Management (SIEM) Reviews and Ratings" [16], the top 15 SIEM platforms are mostly proprietary, with only two open-source options (Elastic Security and Graylog) featured on the list. Among them, only Elastic Security is included in Gartner's 2024 Magic Quadrant for SIEM [17], highlighting its maturity, scalability, and strong community support.

This paper outlines the need for a lightweight, modular, and open-source SIEM solution, specifically designed for vehicles. To address this gap, we propose a system based on Elasticsearch and Kibana, integrated via a Go-based backend interface, to provide data ingestion and visualization of vehicular security events. Our contribution is distinguished by a hybrid approach that combines enterprise-grade SIEM functionality with the inherent limitations imposed by vehicular environments (e.g.: limited computational resources and real-time processing restrictions).

Our hypothesis is that it is, indeed, possible to develop this system that bridges the gap between typical SIEM platforms and the unique limitations of vehicle systems.

IV. DESIGN AND IMPLEMENTATION

A. System architecture

The proposed V2X SIEM system employs a modular, layered architecture designed to handle unique challenges of vehicular networks. Figure 1 illustrates a high-level overview of the system's components.

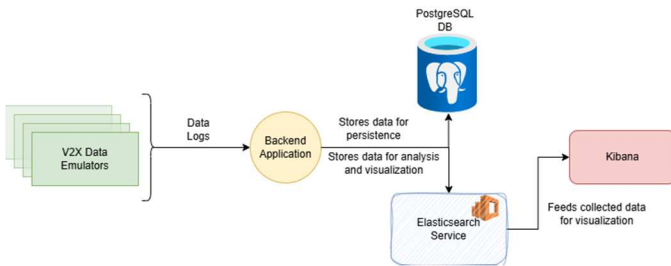


Fig. 1. High-level overview of the proposed V2X system.

The architecture consists of five distinct layers:

- **Collection Layer:** Responsible for capturing V2X communications from multiple protocols (DSRC and C-V2X stack), this layer serves as the primary data ingestion point. It utilizes specialized collectors that parse raw binary messages into a structured format.

- **Processing Layer:** Once collected, messages are processed through a pipeline of security verification and anomaly detection. This layer extracts meaningful security insights from the raw data.

- **Analysis Layer:** The core intelligence of the system resides in this layer, where the rule engine evaluates security events against predefined rules to identify potential threats.

- **Storage Layer:** A dual-storage approach combining PostgreSQL for relational data and Elasticsearch for search and analytics provides flexibility in data persistence and retrieval.

- **Presentation Layer:** Through the integration of Kibana, feeding of the data stored in Elasticsearch, this layer delivers security insights for analytics in an easily interpreted visual display.

The entire solution is containerized using Docker and orchestrated with Docker Compose, allowing for easy deployment and scaling. Among the tools most commonly used in the literature to build a SIEM—namely the ELK Stack, Wazuh, and Splunk—we decided to implement the ELK Stack due to its open-source nature, in contrast to Splunk's proprietary licensing. Additionally, Elasticsearch, which forms the core of the ELK Stack, demonstrated its superior performance compared to OpenSearch, which is the tool used by Wazuh, since Elasticsearch outperforms OpenSearch by 40-140% using 37% less storage resources [18]. The Table I below provides a comparison of the tools to further support our decision to adopt the ELK Stack for this implementation.

TABLE I. SIEM TOOLS CONSIDERED

Feature	ELK Stack	Wazuh	Splunk
License	Open source	Open source	Proprietary
Core Engine	Elasticsearch	OpenSearch	Splunk's engine
Storage Efficiency	High (better than OpenSearch)	Moderate	High (optimized for enterprise use)
Ease of Configuration	Moderate (requires manual setup)	Easy (preconfigured with security modules)	Easy (setup GUI-driven)
Community Support	Large and Active	Growing (security focused)	Large (commercial-focused)

Table II summarizes the key components and their respective responsibilities within the architecture.

TABLE II. KEY ARCHITECTURAL COMPONENTS

Component	Layer	Responsibility
DSRC Collector	Collection	Captures and Parses DSRC protocol messages
C-V2X Collector	Collection	Captures and parses Cellular V2X messages
Security Verifier	Processing	Validates message authenticity and trustworthiness
Anomaly Detector	Processing	Identifies abnormal patterns in V2X communications
Rule Engine	Analysis	Evaluates events against security rules
PostgreSQL	Storage	Stores structured relational data
Elasticsearch	Storage	Indexes events for search and analytics
REST API	Presentation	Provides programmatic access to the system
Dashboard	Presentation	Visualizes security events and metrics

B. Implementation Details

The implementation of the V2X SIEM system utilizes Go (Golang) as the primary programming language, chosen for its performance characteristics, strong typing, and concurrency support—all critical factors for processing high-volume, real-time V2X data. The system's implementation follows a microservices-oriented approach, with each component focused on specific functionalities.

In Figure 2 we can see how the key inner workflows of the application happen in terms of data ingesting and processing.

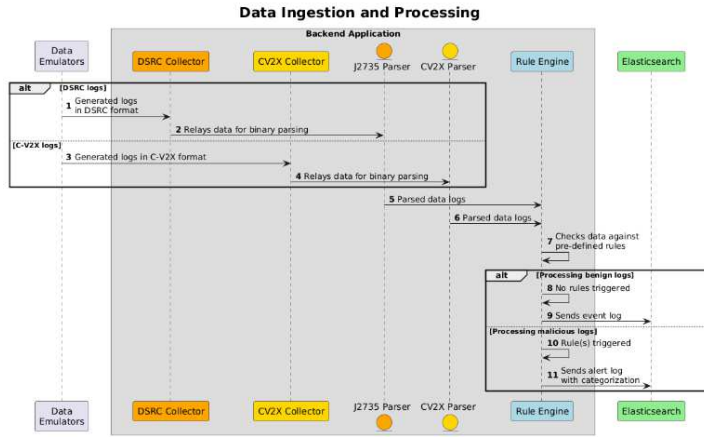


Fig. 2. Inner workflows of V2X SIEM system

1) *Collection subsystem*: The collection subsystem implements specialized collectors for both DSRC and C-V2X protocols. Each collector maintains a UDP socket listening on a configured port. When a message arrives, it is passed to the appropriate parser which extracts structured data according to the protocol specifications. For DSRC, the system implements J2735 standard parsing using a custom ASN.1 decoder that extracts vehicle position (in 1/10 microdegree units), speed (0.02 m/s resolution), and heading (0.0125 degree precision) from binary message structures. C-V2X messages follow multiple formats including CAM, DENM, and SPaT, processed through protocol-specific parsers that handle both J2735-encoded and native C-V2X message formats. Our implementation uses Go's binary encoding package for efficient parsing rather than external ASN.1 libraries, enabling direct field extraction and real-time processing without the overhead of complex ASN.1 parsing frameworks.

2) *Message Processing Pipeline*: Once collected, messages undergo a processing pipeline that includes:

- **Parsing**: Converting binary data into structured objects.
- **Security Verification**: Checking message authenticity.
- **Anomaly Detection**: Identifying suspicious patterns which may include:

- 1) Position jumps (vehicles suddenly appearing in different locations)
- 2) Speed anomalies

3) Message frequency anomalies (flooding attacks)

4) Conflicting information from different sources

3) *Rule Evaluation Engine*: The rule engine forms the analytical core of the system, evaluating security events against predefined rules to identify potential threats. The implementation supports complex rule conditions with logical and comparison operators. Rules are defined with conditions that can reference any field in the security event, including nested JSON structures (post-parsing) and message-specific fields. This flexibility allows for precise threat detection tailored to V2X environments. The system implements comprehensive security rules specifically designed for V2X attack detection. Examples include: (1) **Position_Jump Detection**: which identifies vehicles with unrealistic position changes; (2) **Invalid Signature Detection**: to identify compromised message authentication; (3) **Message Flooding Detection**: for the detection of DoS attacks; and (4) **Speed Anomaly Detection**: to identify implausible vehicle dynamics. These rules leverage the structured parsing of V2X message fields and anomaly detection metadata to achieve precise threat identification.

4) *Data Storage Strategy*: The system employs a dual-storage approach:

- **PostgreSQL Database**: Stores structured data with relational integrity, including:

- 1) V2X messages and their type-specific details (BSM, SPaT, etc.)
- 2) Security events and alerts
- 3) Configuration data (rules, collectors, etc.)

- **Elasticsearch**: Indexes security events and alerts for:

- 1) Full-text search capabilities
- 2) Advanced analytics and aggregations
- 3) Time-based queries and visualizations

This hybrid approach balances the need for relational integrity with the requirements for fast search and analytics over large volumes of data.

5) *API and Dashboard Integration*: The system exposes a comprehensive REST API for programmatic interaction, with endpoints used for either DSRC or C-V2X based protocols, emulated over UDP as aforementioned. For visualization, the system integrates with Kibana to provide rich, interactive dashboard display as can be seen in Figure 3.

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